Fundamentalist vs Mainstream Readings of the *General Theory*:

What Difference Does It Make?

Stephen A Marglin

The theoretical differences are stark. Fundamentalists—I count myself in this group—understand Keynes to have discovered a deep flaw in capitalism. Markets, even perfectly competitive markets, are unable to efficiently coordinate the myriad decisions about production, employment, investment, consumption that agents make. The most alarming manifestation of this failure—the inspiration for the *General Theory*—was the unemployment that characterized the Great Depression.

Most economists consider unemployment to be a short-run problem—despite the lingering effects of the Great Depression for a decade, until the outbreak of hostilities that would become World War II and despite the lingering effects of the slow recovery from the havoc caused by the financial crisis of 2008 that kept the average unemployment rate above 7 percent for the two terms Barack Obama served as President.

We fundamentalists have absorbed a broader lesson from the *General Theory*: the market may not efficiently coordinate economic activities in the long run either. Hence Keynes's call, sketchy as it was in details, for the "socialization of investment."

The mainstream view rejects the idea of a fundamental flaw in capitalism. The problem lies not in the very heart of the system but in departures from the competitive model, in frictions and rigidities, in the exercise of market power. And worse: in government interventions, whether these interventions take the form of a social safety net intended to mitigate the excesses of capitalism or regulatory capture intended to consolidate these excesses.

This essay explores the differences between these two readings of Keynes. At the level of theory to begin with, turning then to the practical differences, the consequences for public policy and its limits in directing the economy.

Two Keynesian Theories

Long before Keynes, long before the Great Depression, the economics profession had recognized a potential problem in the circular flow from production to incomes and back to spending on that production. If the only thing agents can do with their incomes is to buy goods and services, all is well and good. The purchasing power created by the wages, profits, and rents associated with production will flow smoothly back through the market to producers.

Not so once we leave this model for a more realistic—but still drastically simplified—economy in which the magic of money and credit makes it possible for some agents to save, to set aside a portion of their incomes for the future, without committing to the purchase of specific goods to satisfy this goal. And for other agents to spend more than their incomes in order to add to the physical capital stock. In the hopes of profiting from future production and sales, they borrow to finance this investment, promising to repay lenders once these profits materialize.

If investment desires exceed saving desires, total expenditure will exceed total income and goods will fly off the shelves. Production may expand, prices may rise—or both. More worrisome, at least in 1936, was the possibility that investment would fall short, leading to goods piling up on these same shelves. Not far from the minds of readers of the *General Theory* in 1936 was the glut that depressed production and put downward pressure on prices from 1929 on.

Before Keynes's *General Theory*, orthodox economics maintained that neither of these possibilities would long endure. Saving, the subtraction from spending, will be just balanced by investment, the addition to consumer spending. This theory, going back to Adam Smith, held that rational savers, faced with a choice between stashing their saving under mattresses and lending their saving at interest, would obviously prefer the second. In the Smithian view the borrowers are investors engaged in building up the stock of physical capital. The outcome: a lively market bringing lenders (savers) together with borrowers (investors). Here's Smith:

> Whatever a person saves from his revenue he adds to his capital, and either employs it himself in maintaining an additional number of productive hands, or enables some other person to do so, by lending it to him for an interest, that is, for a share of the profits. As the capital of an individual can be increased only by what he saves from his annual revenue or his annual gains, so the capital of a society, which is the same with that of all the individuals who compose it, can be increased only in the same manner. . . What is annually saved is as regularly consumed as what is annually spent, and nearly in the same time too. . . That portion which [an individual] annually saves . . . is consumed by labourers, manufacturers, and artificers, who re-produce with a profit the value of their annual consumption. (1937 [1776], p. 321)

Where there's a market, there's a price, and variations in price adjust supply and demand. If investment demand exceeds the supply of saving, the interest rate will rise, encouraging saving and discouraging investment. And vice versa: if investment falls short of what the community wishes to save, so that expenditure falls short of income, the interest rate will fall. In short, the price mechanism, interest-rate adjustments, insures that saving and investment will find an equilibrium, and expenditure will mirror income.

Adam Smith's view lives on. Here are three examples. The first, from a textbook co-authored by Robert Frank and Ben Bernanke, is noteworthy because its second author is not only a prominent macroeconomist. Bernanke was the Chair of the Federal Reserve when it was faced with the worst financial crisis since the Great Depression. The Frank-Bernanke text compares the balancing of saving and investment with the balancing of demand and supply in the market for apples:

> Desired saving is equated with desired investment through adjustments in the real interest rate, which functions as the "price" of saving. The movements of the real interest rate clear the market for saving in much the same way that the price of apples clears the market for apples. (Frank and Bernanke, 2007, p. 639)

The second example is from the leading elementary text, measured by sales, authored by Greg Mankiw, also a prominent macroeconomist, former chair of the President's Council of Economic Advisers under George W Bush, and the head of the introductory course at Harvard for many years:

> The interest rate in the economy adjusts to balance the supply and demand for loanable funds. The supply of loanable funds comes from national saving, including both private saving and public saving. The demand for loanable funds comes from firms and households that want to borrow for purposes of investment. (*Principles of Economics*, 9th ed., p 538)

Finally, we have Larry Summers, former chief economist at the World Bank, Secretary of the Treasury under Bill Clinton, and top adviser to Barack Obama:

> Just as the price of wheat adjusts to balance the supply of and demand for wheat, it is natural to suppose that interest rates—the price of money—adjust to balance the supply of savings and the demand for investment in an economy. (2016, p. 3)

This idea, with its important corollary that expenditure and income can never stray far from one another, has been enshrined in the economics literature as Say's law, after the 19th century French economist, Jean-Baptiste Say. Say may or may not have believed in Say's Law—much ink has been used on both sides of the argument—but his law is alive and well, to this day the backbone of the mainstream interpretation of Keynes as a theorist of sand in the wheels.

Fundamentalists believe that a principal innovation of the *General Theory* was its rejection of this line of reasoning. According to Keynes, interest rates do not equilibrate saving and investment, which are *flows* of capital, because they are busy elsewhere, equilibrating demand and supply in markets for financial assets—representing *stocks* of capital. In a simplified model in which the only financial assets are cash and long-term bonds,

> The rate of interest is not the "price" which brings into equilibrium the demand for resources to invest with the readiness to abstain from present consumption. It is the "price" which equilibrates the desire to hold wealth in the form of cash with the available quantity of cash. (The *General Theory*, p. 167)

The edifice of the *General Theory* is based on Keynes's answer to the question this view forced upon him: if interest rate(s) regulate asset markets, equilibrating liquidity preference with the opportunity costs of foregoing the returns on illiquid assets, what is the mechanism for equilibrating saving and investment? Keynes's answer: output. Production would increase if a positive shock to demand (or a negative shock to supply) made goods disappear from the shelves; production would fall if goods piled up.

Mainstream Keynesians share the view that Keynes was onto a problem, but with an important qualification: cross out the phrase "even perfectly competitive markets" in thinking about the limits of a market system as coordinating device. Wage rigidity in the face of massive unemployment was proof enough of the lack of competition in labor markets and a symptom of a more general falling away from textbook precepts of perfect competition. Paul Samuelson, one of the first recipients of the Nobel prize after it was extended to economics and a pre-eminent architect of the mathematical turn in mid-20th century economics, put it this way:

> We always assumed that the Keynesian unemployment equilibrium floated on a substructure of administered prices and imperfect competition. I stopped thinking about what was meant by rigid wages and whether you could get the real wage down; I knew it was a good working principle, a good hypothesis to explain that the real wage does not move down indefinitely as long as there is still some unemployment. (Colander and Landreth, pp 160-161).

Ninety years has not changed this view. Introductory economics courses, as taught for four decades at Harvard by Martin Feldstein, Greg Mankiw, and in its current incarnation by Jason Furman and David Laibson, blames market imperfections for the existence of unemployment. Mankiw's elementary text lists three reasons "(1) sticky wages, (2) sticky prices, (3) misperceptions about relative prices." (Mankiw, 9th ed., p 701). The present text, co-authored by Professor Laibson, Daron Acemoglu, and John List, contrasts the effect of a fall in aggregate demand in a regime of rigid wages with the effect of the same fall in demand in a regime of flexible wages:

> With flexible wages, a shift to the left in the labor demand curve reduces the equilibrium wage and employment… With a downward rigid wage, the same leftward shift has a larger impact on employment… because none of the impact… is absorbed by the wage, which remains at its original (rigid) level. Moreover downward wage rigidity causes unemployment: because the wage does not change, the quantity of labor supplied remains the same, but the quantity of labor demanded falls [more than when the wage falls]. (Acemoglu, Laibson, and List, 3rd ed., p 226.)

There are two theoretical issues to deal with. First, why would aggregate demand ever fall in the mainstream story? If investment demand falls, or consumption demand falls and the supply of saving increases, then—assuming Say's Law holds—interest-rate adjustments should guarantee that desired saving and desired investment still balance, and consequently expenditure and income balance, without any changes in aggregate demand.

Putting this issue of logical consistency to one side, there is an inherent ambiguity in the distinction between flexible and rigid wages. Do we mean flexible *nominal* wages or flexible *real* wages? The wage bargain normally sets nominal wages—unless a cost-of-living clause indexes wages to prices. But what presumably matters to agents—capitalists and workers—is the real wage.

Moreover, the advantage mainstream economists give to flexible wages over fixed wages—the soft landing at a new full-employment equilibrium after a negative demand shock—depends on the implicit assumption that a flexible nominal wage translates into an equally flexible real wage. In other words, that money wages can fall without any impact on prices.

My book, *Raising Keynes,* reframes the issue of flexibility in terms of dynamic adjustment to shocks to demand (or to supply). Once we are willing to violate Say's Law and admit these shocks, a smooth landing at a lower level of full employment *à la* Acemoglu, Laibson and List is theoretically possible, but only as a limiting case. The general case is that in a world of perfect competition—with no rigidities, no frictions, no market power—a negative shock to aggregate demand leads to both wages and prices falling, the first in response to unemployment, the second as an attempt to stimulate demand and thereby increase profits. An equilibrium may be reached at which wages and prices are falling at the same rate, so that the real wage is stationary, but in general this equilibrium will be characterized by continuing unemployment. Falling wages frustrate capitalists' attempts to increase sales and production by reducing prices—and vice versa. Stationary real wages are the outcome of a dynamic process, not a built-in rigidity.

The limiting case, in which a flexible nominal wage brings about full employment, requires wages to fall infinitely faster than prices, but this can only happen if we assume rigid prices along with flexible wages (Marglin, *Raising Keynes,* pp). Limiting cases are useful for understanding theory, but as a practical matter, flexible money wages coupled with rigid prices strains credulity.

Even if we put credulity to one side, the full employment of the soft-landing equilibrium is achieved by reducing real wages and making work less attractive. Nor is maintaining full employment by making it less worthwhile to work a benefit for capitalists: the new equilibrium means idle capacity, fewer workers engaging with the same capital than before demand fell. The result is that capitalists could make more money by increasing output but are prevented from doing so by the assumed price rigidity. A restoration of demand would improve matters all around, just as in the general case when falling wages and prices coexist with unemployment and idle capacity.

This leaves the empirical question: are nominal wages actually rigid? The obvious test case is the Great Depression, the most dramatic and severe downturn the capitalist world has experienced and the shock that inspired the *General Theory*. Contrary to the mainstream view, nominal wages moved down dramatically in the United States, as Table 1 shows.

Table 1. Money and Real Wages in the US, 1929-1933

	Avg Annual Earnings of Full Time Employees (Series D 722, p 164)	Avg Hourly Earnings, All Manufacturing (Series D 802, pp 169-170)	Avg Annual Earnings of Full Time Employees, 1929 = 100 (Series D 722, p 164)	Avg Hourly Earnings, All Manufacturing, 1929 = 100 (Series D 802, pp 169-170)	Consumer Price Index, 1929 =100 (Series D 727, p 164)	Avg Real Annual Earnings of Full Time Employees (Series D 722 and D 727, p 164)	Avg Real Hourly Earnings, All Manufacturing (Series D 802, pp 169-170, and D 727, p 164)
1929	$1,405	$0.56	100.00	100.00	100.00	100.00	100.00
1930	$1,368	$0.55	97.37	98.21	97.37	100.00	100.87
1931	$1,275	$0.51	90.75	91.07	88.65	102.37	102.73
1932	$1,120	$0.44	79.72	78.57	79.64	100.10	98.66
1933	$1,048	$0.44	74.59	78.57	75.37	98.97	104.25

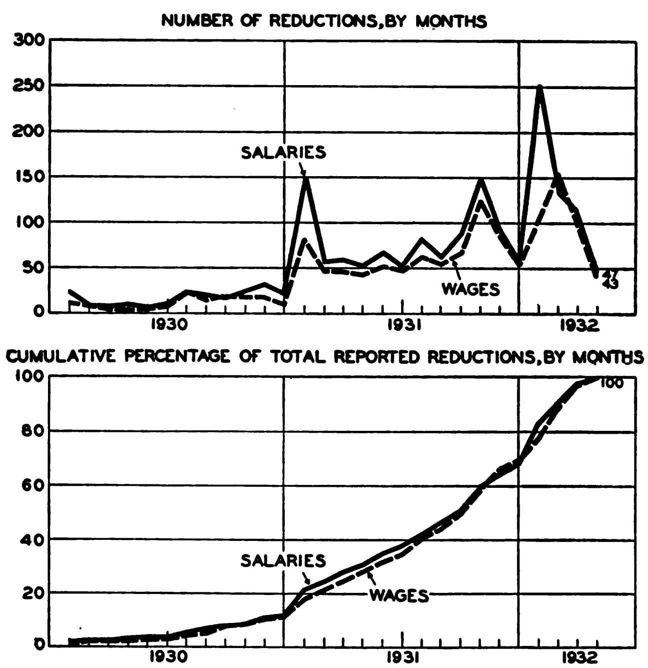
Source: *Historical Statistics of the United States, Colonial Times to 1970*, Part 1

And so did prices. Real wages remained pretty much stationary, so workers who managed to hold on to fulltime jobs were about as well off in 1932 as they had been in 1929, at least in terms of what their average wages would buy—just as the dynamic model developed in *Raising Keynes* suggests. However, if the "average" worker was in debt because of a mortgage taken out to buy a house when he was earning $1400, and had to pay back the loan out of wages that had fallen to $1100, he was in trouble: even if the interest rate declined (a doubtful proposition) the principal payment did not budge even though real-estate values plunged along with the price level.

Like many myths, nominal wage rigidity has a basis in fact. In the Great Depression, reducing wages was a last resort for many employers. The initial response to the fall in aggregate demand focused on output. Prices and wages fell by only a modest 2.5 percent in 1930, while non-agricultural business output fell by 10 percent and employment by 5 percent. But after 1930 prices and wages joined output and employment in a free fall. In 1931 real output was 20 percent below its 1929 levels and employment 13 percent, the same percentage drop as the price level. Things began to improve only with the economic regime change put in place by the Roosevelt Administration in 1933.

A survey by the National Industrial Conference Board in 1932 documented the belated embrace of wage cuts up to the spring of that year.

CHART 5: REDUCTIONS IN SALARIES AND WAGES, JANUARY, 1930, TO APRIL, 1932, BY MONTHS



Source: National Industrial Conference Board, *Salary and Wage Policy in the Depression*, 1932, p 66.

Figure 1

From Theory to Practice

Clearly, mainstream and fundamentalist Keynesians disagree on the mechanisms by which macroeconomic shocks disrupt the economy in the short run. There is much less disagreement on the remedy. Both schools acknowledge the limited role of interest-rate policy in combatting major shocks and embrace the use of fiscal policy, including deficit spending, as the weapon of choice.

This is a relatively recent development. In the immediate aftermath of the Great Recession, the mainstream of the economics profession was hardly convinced that deficit spending could be beneficial. As the Obama stimulus package was taking shape in January 2009, an advertisement placed by the conservative Cato Institute in the *New York Times* and the *Wall Street Journal* on January 28, 2009, had as its central message "we the undersigned do not believe that more government spending is a way to improve economic performance." This ad was signed by some two hundred economists, including several Nobel Laureates (and future Laureates), http://object.cato.org/sites/cato.org/files/pubs/pdf/cato_stimulus.pdf (accessed February 2, 2016).

After the stimulus did its work, a panel of prominent economists interviewed by the University of Chicago's Booth School of Business in 2012 was close to unanimity that the Obama stimulus had indeed created jobs, but barely half agreed that the game was worth the candle. By 2014 the profession had moved towards a consensus: almost two-thirds (of a slightly larger Booth panel) concurred with the conclusion that the benefits of the stimulus exceeded the costs.

There is by now broad bipartisan agreement in the US as well. Opposition to the Obama stimulus was the last stand of traditional Republicans against countercyclical fiscal policy; the Republican Party, remade in the image of Donald Trump, was quick to endorse massive deficits to counter the economic impact of the Covid epidemic in 2020, and Joe Biden and a Democratic-controlled Congress continued deploying the budget to this end in 2021.

If there is agreement on the remedy, does it matter that views differ as to the causes of the problems? Yes, because the agreement is very limited: mainstream economists, for the most part, are willing to concede that in the short run, frictions, rigidities—warts on the body of capitalism—may require government intervention, but these same economists, including those who identify as mainstream Keynesians, argue that there is no constructive role for the government in the long run—except to insure that inflation is limited to a modest rate (2 percent is the magic number in the policy world). The warts disappear if we just wait long enough.

The Long Run

With the warts gone, aggregate demand no longer matters. Or, rather, aggregate demand has no influence on the real economy, on output and employment, on growth of production. Aggregate demand can only influence the price level—hence the emphasis on controlling inflation.

This can be seen in vertical Phillips curves and (what amounts to the same thing since derived by the same logic) vertical aggregate-supply curves that populate elementary texts. And even more so in the discussions of the sources and drivers of long-run growth. In particular, capital accumulation is dismissed as a long-run driver of output growth on the grounds that capital accumulation will—presumably sooner rather than later—run into the headwinds of decreasing marginal productivity (Acemoglu et al, ch 7, esp. p 157). Similarly, on the supply side education is relegated to the basket of short-run improvements that will peter out in the long run.

Fundamentalist Keynesians have long opposed the mainstream view of the long run as the province of the supply side. My own fundamentalist effort, *Raising Keynes,* concludes with a theory of the long run in which aggregate demand matters for the real economy as well as for the price level.

How do we go about deciding between the two views? The first question to ask is how long is the long run. The long run can be taken as any period over which the capital stock varies, distinguishing it from a short period in which the capital stock is assumed to be fixed. This makes for a very short long run.

At the other extreme, the long run can be understood as a period lasting until the end times, an infinity away. In one sense this definition is more favorable to orthodoxy, for even the most committed fundamentalist must recognize that supply-side considerations will eventually dominate. But the supply side in question is one in which the sun has succumbed to old age and entropy has made survival impossible, hardly the supply-side of the mainstream imagination.

For practical purposes, the focus of this essay, it makes sense to think of the long run as neither the day after tomorrow nor the infinite future, rather in terms of something like a generation, long enough for very different forces to be at work than the forces that dominate the here and now, but short enough so as not dissolve in the mists of an unforeseeable future. Over this period, I will argue, demand clearly matters.

How Did Mainstream Economists Come to Believe Demand Doesn't Matter?

The mainstream position took shape in the years following the publication of the *General Theory,* beginning with Roy Harrod's attempt to draw out the implications of Keynes's reasoning for the likely course of capitalism over time (Harrod, 1939; 1948). Harrod's argument assumed that the forces governing equilibrium between entrepreneurial sentiment and the community's propensity to save, the "warranted rate of growth" of output, were separate and distinct from the forces governing the "natural rate of growth" of the labor force.

If the two rates happened to coincide, the rate of growth of output and employment would reflect this happy coincidence. The more likely outcome is a mismatch between the two rates. What happens in this case depends on how Harrod's theory is translated into a model. One possible assumption is that the capital:labor ratio is fixed; one worker to one machine. Given this assumption, we might expect that capital growth in excess of labor-force growth would lead to constant boom conditions, with capitalists chronically short of workers. By contrast, if the labor force grows more rapidly than the capital stock, we would expect to find unemployment and depression.

Harrod reaches opposite conclusions by way of his assumptions about how the economy adjusts out of equilibrium. Harrod's exposition is somewhat ambiguous, but three additional assumptions allow us to make sense of his counter-intuitive conclusions. One is that investment demand—the demand for additional capital—is a constant fraction of the anticipated increase in output. A second assumption is that at a stationary level of output there would be no incentive for capitalists to add to the capital stock. Along with these assumptions Harrod supposes that the community's propensity to save is a constant fraction of income.

These three assumptions together imply that the equilibrium between investment demand and the supply of saving is unstable: if output is greater than the equilibrium level, investment exceeds saving, and the adjustment process leads to higher output and income—and an ever-widening gap between desired investment and desired saving. Ditto for the opposite case, in which a negative shock reduces output: saving exceeds investment, and output falls. The picture is in Figure 2.

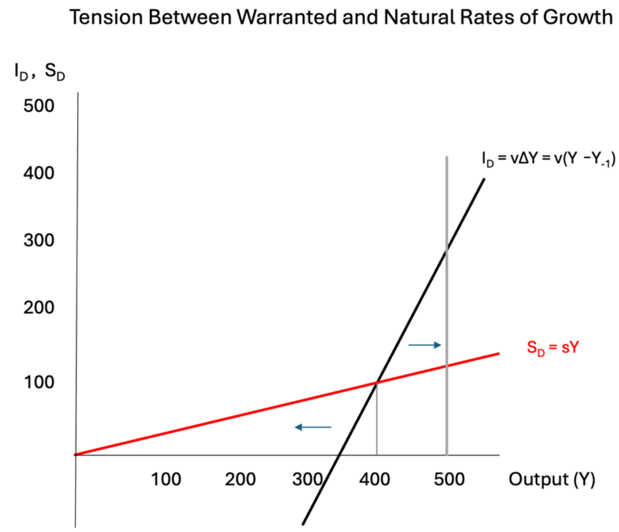


Figure 2

The consequences for the tension between warranted and natural rates of growth is this: at any moment of time, a shortfall of the output dictated by the warranted level of growth, say 400, relative to the level of output corresponding to the natural rate of growth, say 500, produces the possibility of a protracted boom, with output and employment continually pressing against the limit imposed by the rate of labor-force growth. On the other hand, if actual income is below 400, there is nothing to stop collapse, output falling in response to a continual shortfall of investment demand relative to saving.

Suppose that instead of 500, the natural rate of growth corresponds to 300. This would render a protracted boom impossible. Sooner rather than later, an excess of investment demand relative to saving would run up against the limits of a slowly growing labor force, producing inflation rather than real growth. On the other side, if output were to fall below the warranted-rate level, there is no brake that would impede the collapse of the economy; the natural rate imposes a ceiling when the rate of growth dictated by investment and saving is above the warranted rate, but is no barrier when this rate is below the warranted rate.

With or without instability in the out-of-equilibrium relationship between investment and saving, Harrod's model leads to pessimistic conclusions about the possibility for the evolution of a harmonious equilibrium under capitalism. Independent theorizing by Evsey Domar (1946, 1947) reinforced Harrod's conclusion. In fact, Domar's model is very different from Harrod's, but from the get-go a common equation joined the two models together in the literature—despite the difference in the meaning this equation had in the two models. (See *Raising Keynes,* ch 17, for a discussion of the differences.).

Recall that that the discussion around these models took place in the early post World War II period. Given the history of the Great Depression, concern for maintaining full employment was a central question of public policy. The conclusions regarding the precariousness of a full-employment equilibrium provided fuel for the partisans of an active role for government to manage the economy.

At the same time there was considerable confusion about the meaning of the Harrod-Domar equation; the very fact that one equation could have two very different meanings suggests a problem. In this environment, Robert Solow's 1956 paper was a breath of fresh air, seeming to clarify the ambiguity in the literary, discursive presentations of both Harrod and Domar by providing a mathematical model deploying simple calculus. It is fair to say that this essay marked a watershed in the mathematization of economics that took place over subsequent decades.

As Solow laid out the problem

> The characteristic and powerful conclusion of the Harrod–Domar line of thought is that even for the long run the economic system is at best balanced on a knife-edge of equilibrium growth. Were the magnitudes of the key parameters— the savings ratio, the capital-output ratio, the rate of increase of the labor force—to slip ever so slightly from dead center, the consequence would be either growing unemployment or prolonged inflation. In Harrod's terms the critical question of balance boils down to a comparison between the natural rate of growth which depends, in the absence of technological change, on the increase of the labor force, and the warranted rate of growth which depends on the saving and investing habits of households and firms. (1956, p. 65)

Solow's resolution of the knife-edge problem is elegant: instead of fixed coefficients in production, Solow assumes substitutability between capital and labor. He adds to this the assumption that the entire labor force is fully employed, a job for every willing worker. With these assumptions, his model allows capital-stock growth to adapt to the exogenously given rate of labor-force growth. In the long run growth is supply-side determined, by the growth of the labor force alone if technology doesn't change, by a combination of technological change and labor-force growth in a more realistic model.

Along the way the warranted rate of growth changes its meaning. No longer does it represent an equilibrium between investment demand and saving supply. It is now determined by the propensity to save and the capital:output ratio. (In the simplest interpretation of Harrod's model the capital:output ratio is a fixed parameter determined by fixed-proportions technology; for Domar, the multiplier process determines how much output must increase to generate additional saving equal to an exogenously given increase in investment demand.)

For all its elegance in demonstrating the existence of a full-employment path for the economy, Solow's solution falls short. It simply assumes away the Keynesian problem that lay at the heart of Harrod's and Domar's inquiries: the role of aggregate demand, and more specifically, the role of investment demand. In this sense the knife-edge problem is a side-show; the tension between the determinants of the growth of demand and the growth of supply exists with or without the knife edge.

Solow himself was well aware of this. The last section of his 1956 essay begins by acknowledging what is left out:

> Everything above is the neoclassical side of the coin. Most especially it is full employment economics—in the dual aspect of equilibrium condition and frictionless, competitive, causal system. All the difficulties and rigidities which go into modern Keynesian income analysis have been shunted aside. It is not my contention that these problems don't exist, nor that they are of no significance in the long run. (1956, p. 91)

But not only is there nothing in Solow's essay to address Keynes, this essay set mainstream economics along the path it has followed for more than two generations: the long run is the province of an economics in which technology and labor-force growth rule the roost, and demand can only affect the price level.

Theory aside, the facts seem to support Solow's view: booms and busts have occurred, but not the persistent tendency to one or the other that Harrod and Domar predicted. The reasoning however is circular: the absence of persistent boom or bust supports the view of demand adjusting to supply if the supply of labor is indeed exogenously given. But that is an assumption, even if it is an assumption enshrined in mainstream growth theory from the get-go. To this assumption we now turn our attention.

The Supply Side Responds to Demand in the Long Run

In the spirit of W Arthur Lewis's unlimited supplies of labor, I argue that the absence of persistent boom or bust is the result of supply adapting to demand rather than the other way around. That is, the supply of labor is endogenous.

I make an argument for labor-supply endogeneity in *Raising Keynes* and with greater attention to the difference with Lewis's own argument in a contribution to a symposium on Lewis (Marglin, 2018). An essential element of my argument is that the capitalist economy is not *the* economy, that the capitalist economy is embedded in a larger economic system, which allows the capitalist economy to draw on other economies for labor as needed; non-capitalist economies are sources of labor and—exceptionally (as in the Great Depression)—sinks.

Appropriately, theories of growth take account of distinctive features of capitalism such as wage labor, which makes them *ipso facto* theories of *capitalist* growth. It is a category mistake to identify capitalism with the whole economy and assume that a theory of capitalist growth stands in for a universal theory of growth.

How well does the idea of a separate capitalist economy work for the US? Historically there have been two parallel economies which have served as primary reservoirs of labor for the capitalist economy: the family-farm economy and the household economy.

For the better part of two centuries, US agriculture has largely been organized around production for the market, not for subsistence. Does this not make it capitalist? No, because a critical ingredient of capitalist production, wage labor, has played a minor role, serving as an auxiliary for the family in times of peak labor needs, particularly at harvest time and, to a lesser extent, at planting time.

How about the household as a reserve army? In the household economy, goods and services are provided, largely by women—kids are driven to soccer practice or piano lessons, beds made, laundry done, meals cooked—without a penny changing hands. Again, wage labor is not totally absent, but (except in the richest households) nannies, sitters, cleaning ladies, and take-in are supplements to the unpaid labor of mothers (and fathers) rather than the mainstay of the household.

A third reservoir of labor is immigration. In the US, immigration has fueled capitalist development since the beginnings of European settlement. Other countries in which capitalism has taken root, the nations of Western Europe in particular, have more recently come to depend on immigration to backstop farm and kitchen in providing workers to satisfy capitalism's growing demand.

In Marx's terminology, the family farm, the household economy, and immigration are *reserve armies* that allow capitalism to expand beyond the labor resources provided from within the capitalist economy itself. The important point is that reserve armies have not been static, a given number of battalions. Rather, reserve armies are created and recreated to serve the needs of capitalism. When one was exhausted, another grew up in its place.

Case in point: in the middle of the 20th century, as the family-farm population in the US dwindled, women's participation in paid capitalist labor accelerated. Not long after, a decades-long policy of restrictive immigration was reversed, and the percentage of foreign-born Americans climbed back to what it had been at the turn of the 20th century.

The 21st century may be different. The farm-labor force has been reduced to a vanishingly small percentage of the total workforce, and for the most part what remains is a workforce of wage laborers employed by capitalists. Women's rate of participation in the paid labor force is roughly equal to men's. What remains is immigration, and that has turned out to be a flash point in the war that the American government, under Donald Trump, has unleashed on its own country. Despite the claims (and fears) that automation will make the need for a reserve army moot, it seems more likely that growth cannot be sustained without continued reliance on immigrants. Time will tell.

The American experience has been repeated, with a time lag, around the world. The three charts below show the decline in agriculture and the accompanying rise in women's participation rates as well as the proportion of immigrants in the populations of Spain, South Korea and China over the past half century, along with a chart depicting the US experience in the 20th century.

Figure 3

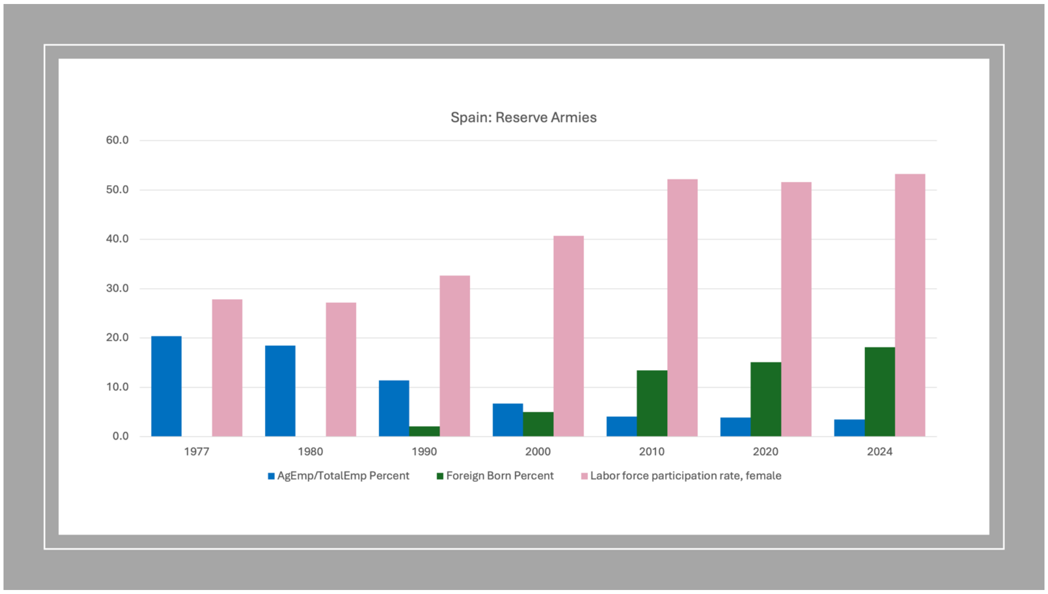


Figure 4

Spain is particularly noteworthy for its 180° turn from being a country spilling out impoverished workers in the early post World War II years to becoming one of the countries in Europe with the largest share of immigrants in its population. Immigration began in earnest in the last decade of the 20th century and accelerated sharply in the early years of this century, as agricultural employment fell below five percent of total employment and female participation in paid employment leveled off. I'll return to this point.

Figure 5

Korea, too, has shown a surprising openness to immigration (albeit restricting paths to citizenship) given its status prior to this century of one of the world's most, if not the most, ethnically homogeneous populations. Observe that in South Korea, women's participation rate leveled off in the 1990s, at the same time that the farm share of employment fell to 10 percent and thereafter had little to contribute to non-farm employment. Not coincidentally, this is when its openness to immigration first began. As it was for Spain, the timing of the switch in reserve armies is relevant to the larger story of Korean growth, supporting the idea of new reserve armies being called into existence as older ones evaporate.

China is exceptional in that substantial reserves of agricultural labor have existed well into the 21st century, with a quarter of employment still in this sector. For this reason, Chinese growth has not been limited by the fact that women's participation, already high a half century ago, had no upward margin to exploit. Nor has China, unlike its East Asian neighbor, welcomed more than an insignificant number of immigrants. Figure 6 provides the picture.

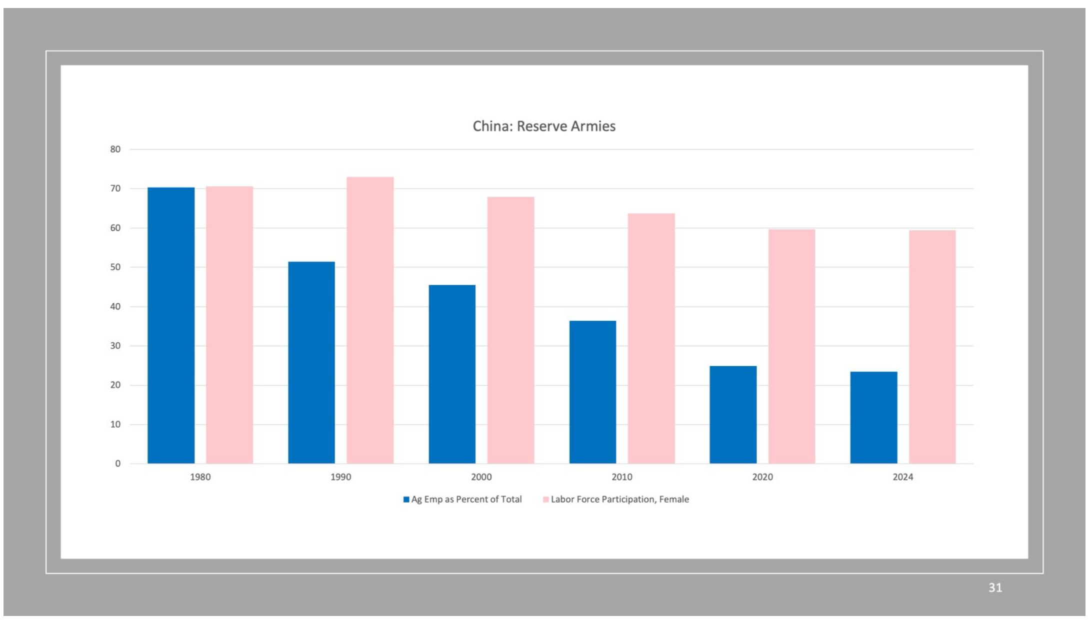


Figure 6

Table 2 documents non-agricultural and total employment-growth rates for all seven countries in my sample.

Table 2: Employment Growth							
	Non-Ag Employment Growth (Annual Percntage Rate)				Total Employment Growth (Annual Percntage Rate)		
	1971-1996	1978-1996	1996-2019		1971-1996	1978-1996	1996-2019
Korea	5.22	4.16	1.40		2.96	2.45	1.14
Spain		0.97	1.98		0.10	0.22	1.78
USA	1.94	1.59	1.00		1.87	1.54	0.95
China	6.79	6.58	1.89		2.15	2.24	0.47
UK	0.29	0.29	1.05		0.25	0.24	1.02
France	0.90	0.74	0.79		0.47	0.41	0.72
Japan	1.40	1.38	0.28		0.94	1.01	0.17

Spain and China both managed close to 2 percent non-agricultural employment growth between 1996 and 2019, by tapping very different reserve armies. Spain saw its agricultural sector shrink to practically nothing in terms of employment by the end of the 20th century, at the same time the percentage of women employed in the paid labor force was rising less rapidly. It embraced immigration to fill the gap. China, by contrast, still had substantial reserves of underemployed peasants in this century, enough to sustain its somewhat reduced demand for workers outside of agriculture.

The pool of farm labor has not played much of a role in mainstream theories of growth, being seen as a one-off transitional phenomenon. In contrast, economic historians and development economists have emphasized the central role of incorporating peasants into the capitalist labor force. In the case of China, for example, Cai Fang has over the years invoked Lewis's theory as a template both for the phase of rapid growth and, in the mid-teens of this century, as a "turning point," for the slowdown in growth as the pool of peasant labor shrunk (Cai Fang, 2006, 2016).

What both groups of scholars have missed is that surplus agricultural labor is just one specific instance of a more general phenomenon. Namely, that reserve armies are created and recreated to serve the needs of capitalist expansion.

A second aspect of the supply side is the quality of the labor force. The importance of education in creating the conditions for sustained economic growth is accepted across the board, by mainstream economists and heterodox economists alike. In this respect fundamentalist Keynesian economists are just part of a general consensus.

Proxied by educational achievement, labor-force quality has shown significant improvement in all seven countries over the last half century. The US, which even 50 years ago had a highly educated workforce, saw high-school completion rise by more than 50 percent as a percentage of the population aged 25 or more, from just over 60 percent in 1970 to more than 90 percent in 2010. In the UK, Japan, and Korea, secondary school completion rose from the single digits in 1970 to 70+ percent in 2010. France did less well, but not by much: its percentages were 10 in 1970 and 61 in 2010. The most dramatic increases were in the countries that started out with high-school graduates constituting less than 5 percent of the 25+ population, China and Spain. In 2010 almost half the 25+ population of Spain had completed a secondary education. China, despite a dramatic increase over the four decades, trailed: only one-fifth of this age group had a high-school degree in 2010 (Barro-Lee, 2014).

Education, though generally understood as exogenous in mainstream models of growth, clearly has an endogenous, demand-driven, component. The considerable expenditure required to provide primary and secondary schools, both the capital expenditure and the continuing outlays on teachers and school supplies, represents a policy choice, not a given.

Then there is the elephant in the room: ecological constraints on the availability of sinks for the waste products of growth and sources of essential raw materials. These are not constraints like the cooling down of our sun that dissolve in the mists of time future. Sinks for the wasteful by-products of growth (think global warming) and sources for capital equipment (think rare earths) are issues of the here and now, issues that certainly are relevant to a time horizon of a generation or two. Indeed, these ecological constraints may be the most purely exogenous constraints on growth, the unadulterated supply side.

The Demand Side: Capital-Stock Growth

The fundamentalist long run includes an important role for the supply side. Different from the mainstream—the "supply" side is to an important extent demand determined—but a part of the story.

Necessary but not sufficient for understanding growth. Ecological constraints may make growth counterproductive to human wellbeing even if growth remains an option. Reserve armies may provide a continuous flow of workers to the capitalist sector, and education will certainly improve the quality of the workforce. But there is no guarantee that these workers will find useful employment. Capital-stock growth is necessary as well.

For the mainstream, the capital stock is only a hiccup on the long-run growth trajectory. The Acemoglu-Laibson-List text is once again instructive:

> Can physical capital accumulation by itself generate sustained growth—where real GDP per capital grows at a positive and relative steady rate for an extended period of time? The answer to this question is "no" for a simple reason: *the diminishing*

marginal product of physical capital. (Acemoglu, Laibson and List, Macroeconomics, *3rd edition, ch* 7.2, p 157, emphasis in original.)

The data tell a different story. In my seven-country sample, the outliers with respect to GDP growth, China and Korea, are also outliers with respect to the growth of the capital stock. Compare Figure 7, which represents GDP growth per worker, and Figure 8, which represents capital-stock growth.

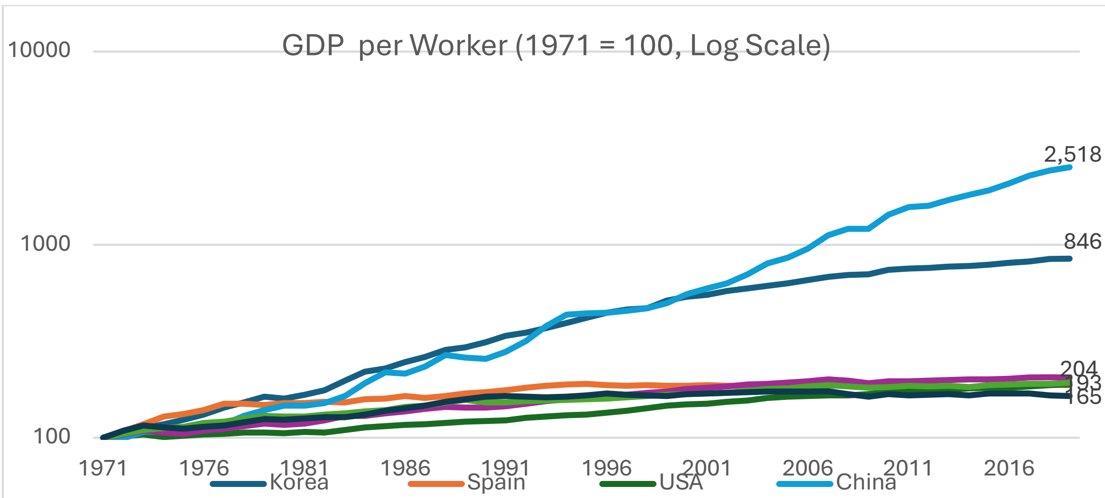


Figure 7

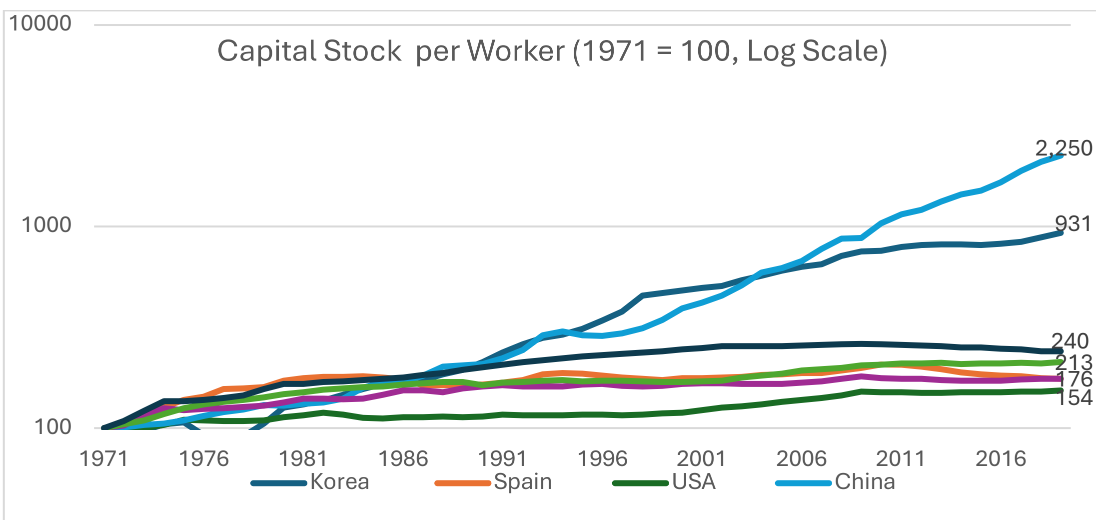


Figure 8

China and Korea move pretty much in lockstep from 1971 to the mid-to-late 1990s; output per worker grows at approximately the same pace as capital per worker. After the Asian financial crisis, the two paths diverge. Korean growth, both of output and capital per worker, falls markedly while China continues its earlier trajectory.

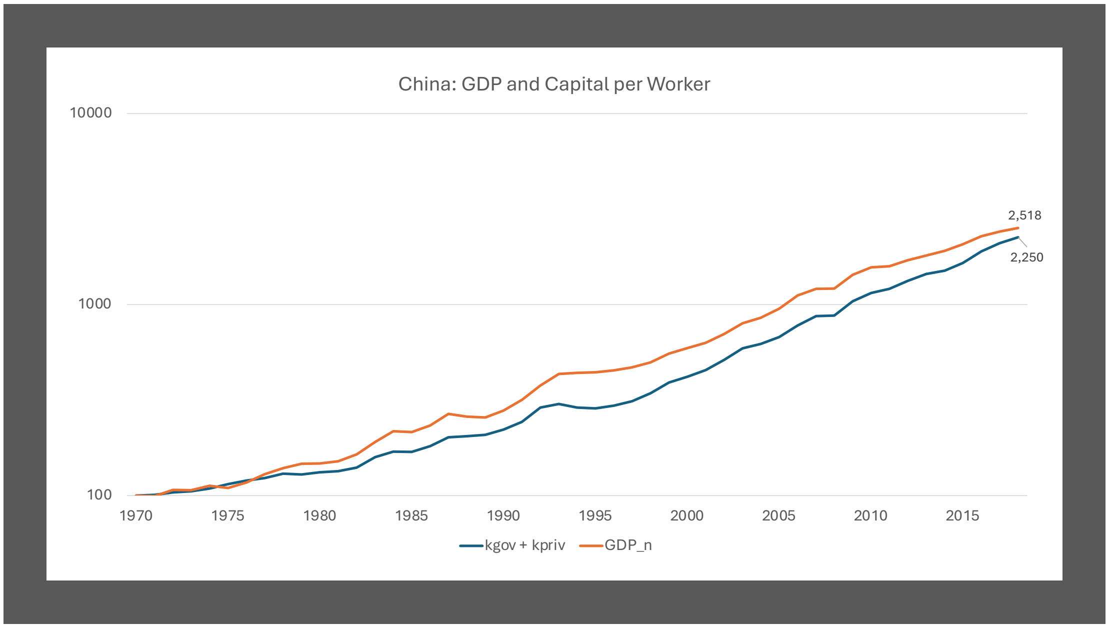


Figure 9

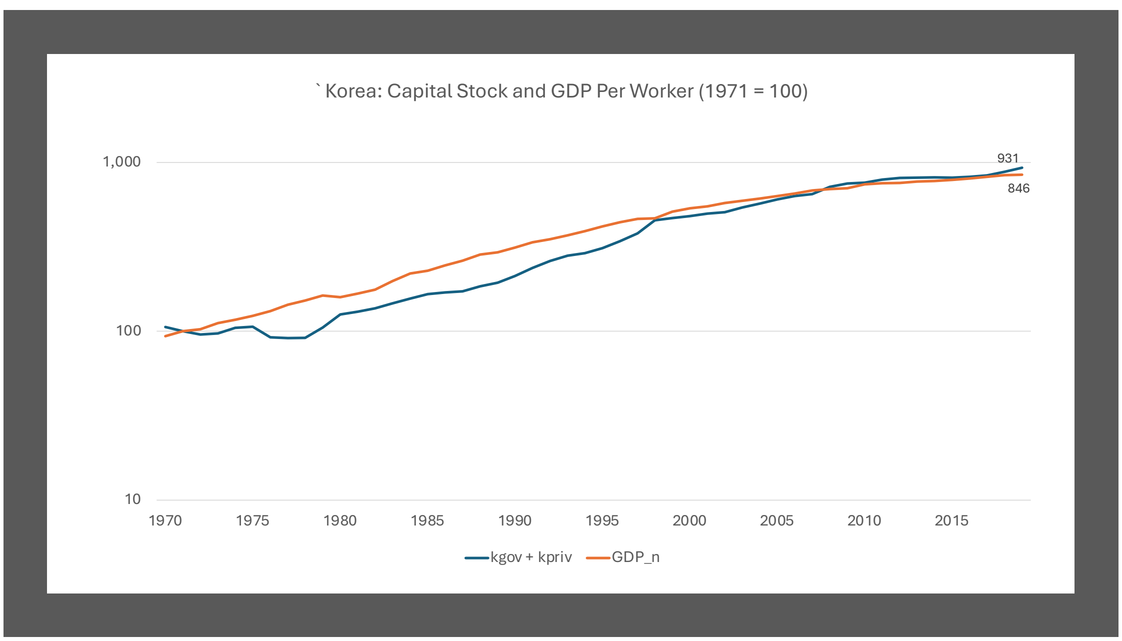


Figure 10

All told there is considerable diversity within my sample. In the US and the UK output per worker outstrips growth in the capital stock, in the US from the mid-1980s, in the UK from the mid'90s, in both by some 20 percent over the half century. Japan is exceptional in the opposite direction, with capital-stock growth consistently higher than output per worker. Here are the pictures for five other countries.

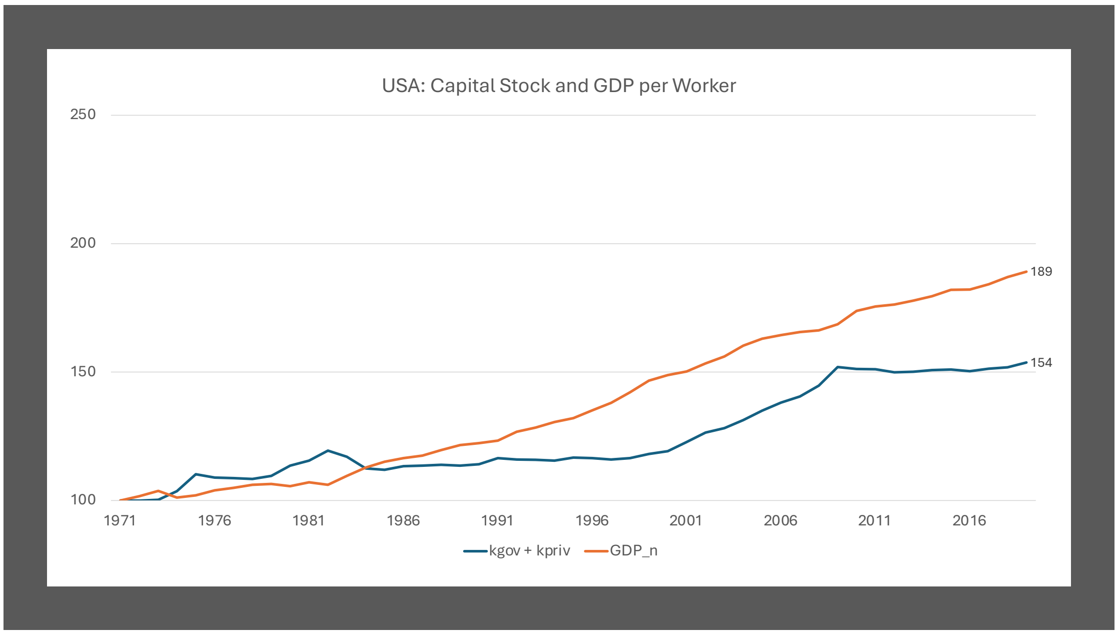


Figure 11

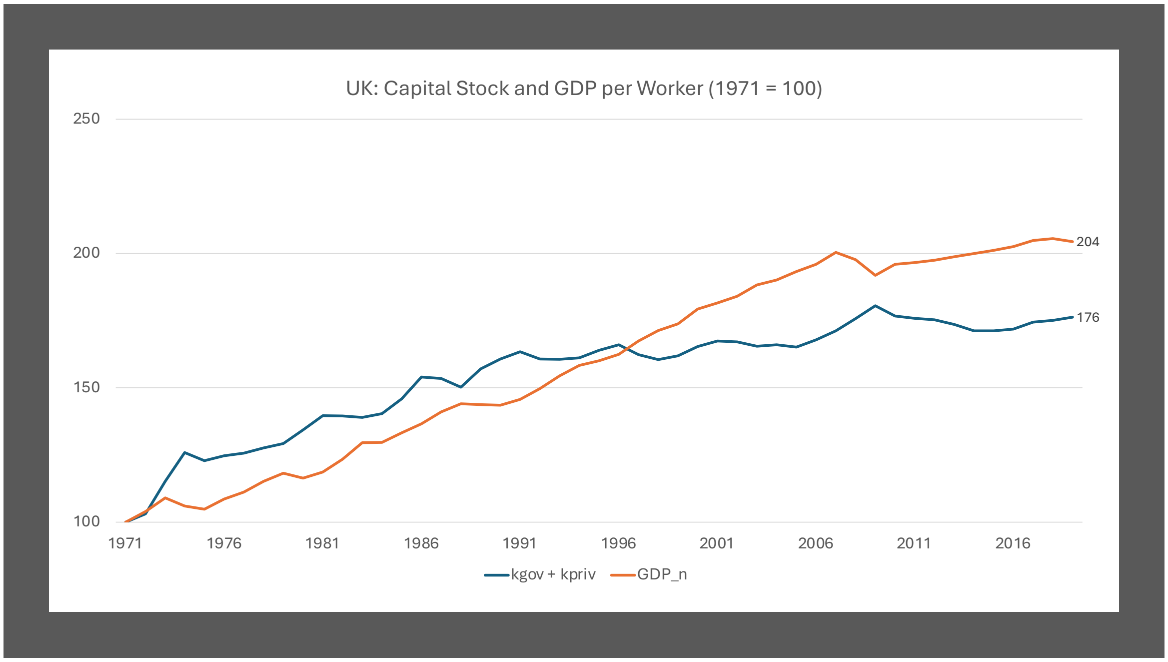


Figure 12

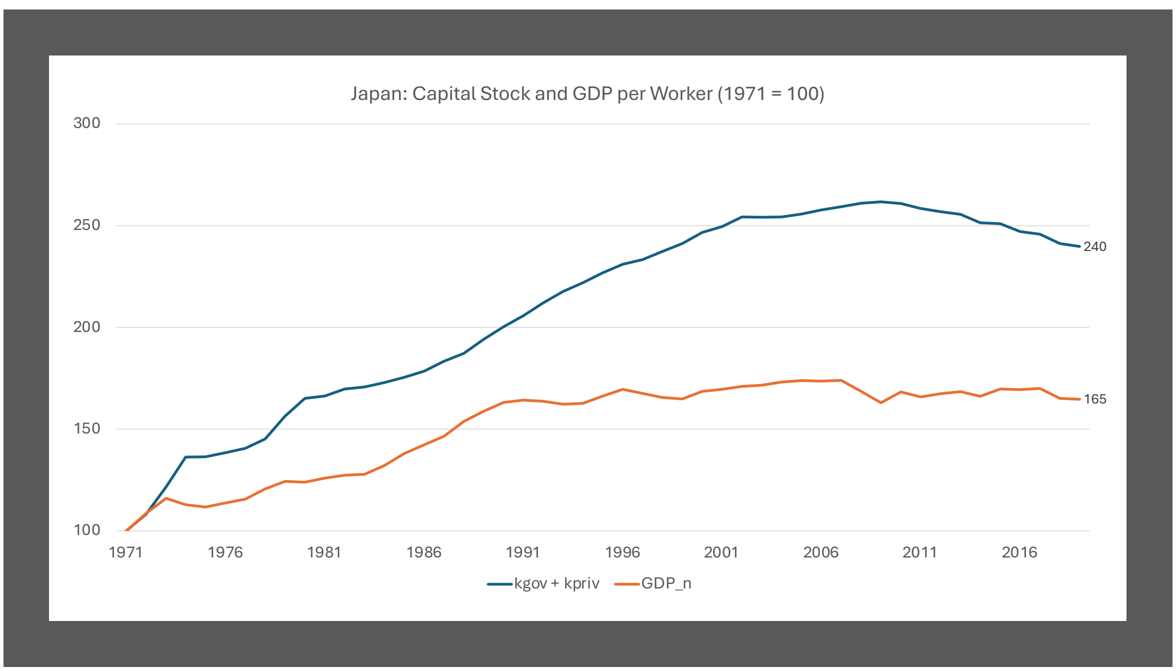


Figure 13

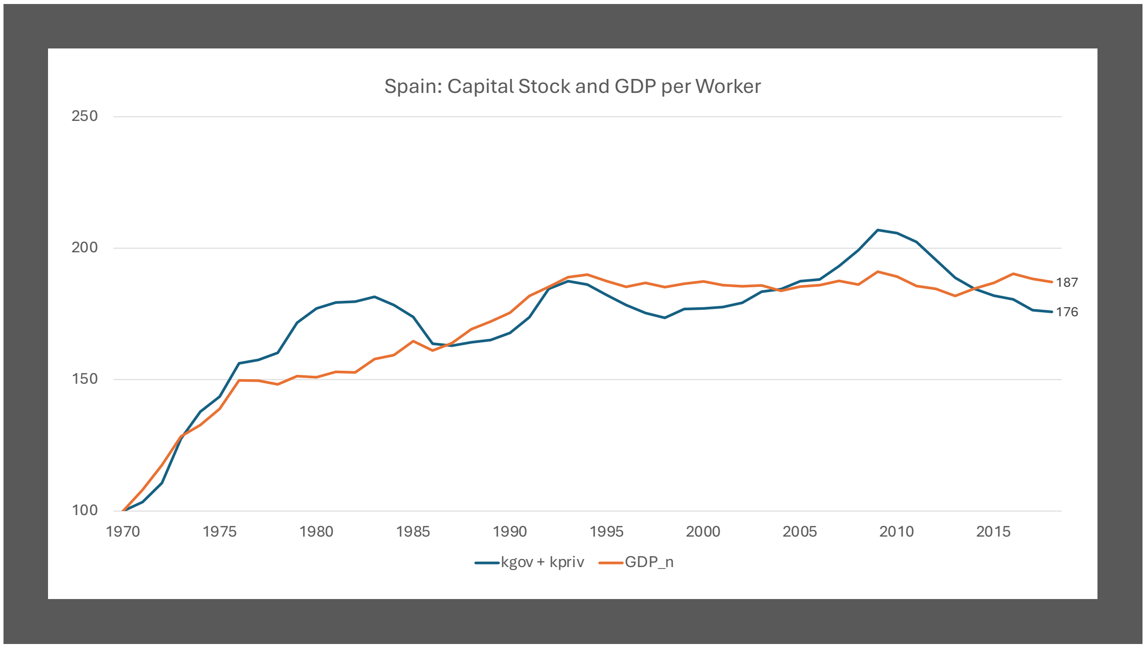


Figure 14

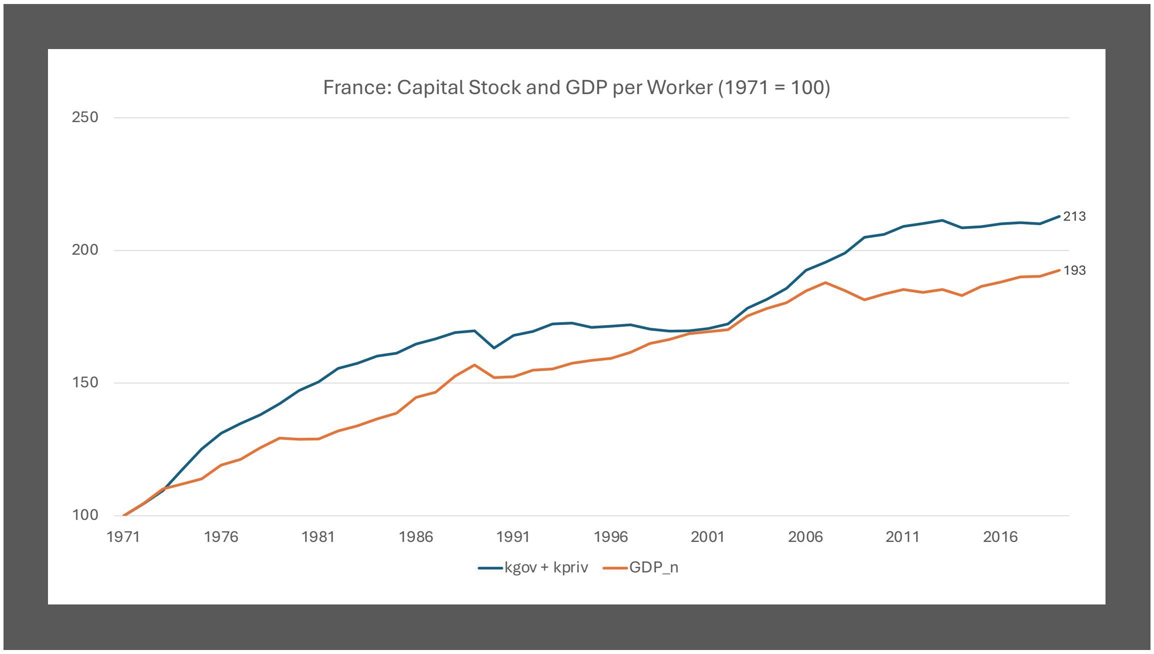


Figure 15

If we look back at what is happening to their respective reserve armies, as shown in Figures 5 and 6, tying the Korean slowdown to the exhaustion of surplus farm labor seems less plausible than the Lewis "turning point" story suggests. From the 1970s to the 1990s, non-agricultural employment grew at an annual rate of 5 percent while total employment grew at close to 3 percent annually. Over the next quarter century, non-agricultural employment grew at the same rate as overall employment, barely 1 percent per year. From 1998 to 2019, capital-stock growth outpaced productivity growth by only 10 percent cumulatively, hardly an indication of capital saturation.

China also experienced a marked decline in the growth rates of both total and non-agricultural employment after 1996—from 2 and 7 percent respectively in the earlier period to less than 0.5 and 2 percent in the later period. Nevertheless, output per worker actually grew more rapidly after 1996 than before. As was the case for Korea, the result was that capital grew faster than output, but, unlike Korea, this did not retard output growth. However important the agricultural reserve army was for China's remarkable 20th century output growth, its depletion did not prevent even stronger growth in the first two decades of the 21st century. In the earlier period, the mobilization of reserve armies contributed to China's ability to defy mainstream reasoning that capital accumulation would sooner rather than later run into diminishing returns. The experience of the two decades of this century that ended with the covid epidemic demonstrated that the pace of output growth could be maintained without the massive influx of erstwhile peasants.

What explains capital-stock growth? The *General Theory* famously highlights the importance of capitalist psychology. One of the oft-most quoted passages likens the undertaking of investment to a voyage to the South Pole:

> [A] characteristic of human nature [is] that a large proportion of our positive activities depend on spontaneous optimism rather than on a mathematical expectation, whether moral or hedonistic or economic. Most, probably, of our decisions to do something positive, the full consequences of which will be drawn out over many days to come, can only be taken as a result of animal spirits—of a spontaneous urge to action rather than inaction, and not as the outcome of a weighted average of quantitative benefits multiplied by quantitative probabilities. Enterprise only pretends to itself to be mainly actuated by the statements in its own prospectus, however candid and sincere. Only a little more than an expedition to the South Pole, is it based on an exact calculation of benefits to come. Thus if the animal spirits are dimmed and the spontaneous optimism falters, leaving us to depend on nothing but a mathematical expectation, enterprise will fade and die. (*General Theory,* pp 161-162).

No doubt. But government policy is important too, most obviously in China but in capitalist economies as well. For one thing the collective share of investment undertaken by governments directly is crucial for overall growth. Even in the United States, bastion of capitalism, the public sector accounted for more than 40 percent of the capital stock in 1971 and almost 30 percent in 2019, by this time approximately the same share as in the other five capitalist economies. (China, though ostensibly socialist, went the other way: private capital went from barely 10 percent of the total in 1971 to more than half of a greatly expanded capital stock in 2019.)

For the private sector government policy is also crucial. The rate of investment, and ultimately the stock of capital, is presented in *Raising Keynes* as a policy choice. In his 1980s book, *Day of Reckoning,* Ben Friedman makes a more orthodox argument that leads to the same conclusion. For any given level of employment and output there exist trade-offs between private consumption, private investment, and collective consumption and investment. For both Friedman and me, the government has two instruments to influence the composition of output, the interest rate and the budgetary stance. The tax rate determines how much disposable income is left in the hands of consumers and consequently the level of consumption, and the interest rate then determines the mix between private investment and government spending. A high tax rate frees up resources by curtailing private consumption, and a low interest rate directs these resources towards private investment.

This oversimplifies and leaves out the many factors that have made for high rates of household and business saving in China and Korea. (Most accounts emphasize the lack of a safety net that leads to high rates of saving for retirement and rainy days.) But the simplification makes an important point, namely, that government policies have important consequences for the level as well as the composition of private capital accumulation. Virtually all accounts of the Chinese and Korean achievement in mobilizing resources for investment emphasize the role of government, though operating in very different political and economic environments and utilizing different tools. Or, rather, utilizing different combinations of tools at different times, from control of credit, determining the volume of investment as well as which firms had more favorable access and lower interest rates, to (in the case of China) the pervasive use of off-budget fiscal policies to stimulate investment.

The Phillips Curve in the Long Run

In stark opposition to mainstream growth theory, economists viewing growth from a fundamentalist Keynesian tradition argue that there is a trade-off operating in the long run as well as in the short run. Higher growth, when set in motion by higher demand, leads to higher inflation—the Phillips curve. *Raising Keynes* explored this in the US context, focusing on separating out shifts due to supply-side shocks from movements along a given Phillips curve due to changes in demand.

Both the shifts along the Phillips curve and the shifts of the Phillips curve are clear in Figure 16, reproduced from *Raising Keynes*.

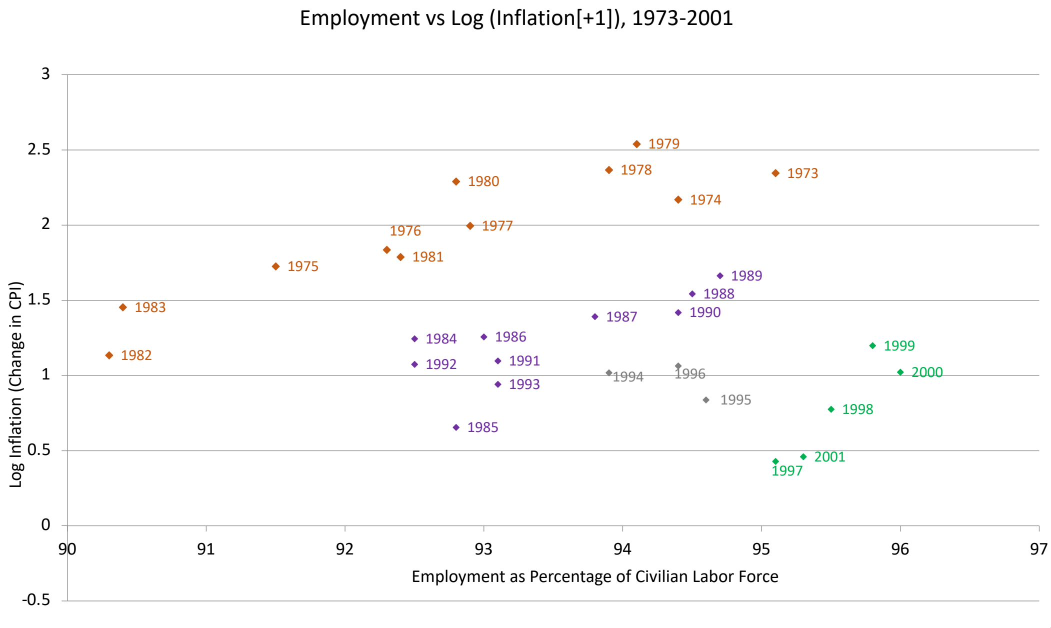


Figure 16

Three distinct Phillips curves are visible. The uppermost one covers the period from 1973 to 1983, the period of the two oil shocks. The middle one reflects the decade from 1984 to 1993, the post-oil-shock period in which baseline inflation expectations, reflected in the constant term in the Phillips equation, fell significantly. The curve in the lower right-hand corner reflects a very different shift, namely, the imposition of a new capital-friendlier regime, in which workers accepted a reduction in the wage share; baseline inflation fell because of less pressure on wages. The three years 1994-1996 represent the transition to this new regime. Interestingly, there is no indication that changes in demand and employment triggered smaller changes in inflation after 1996. On the contrary, the Phillips curve that prevailed in Bill Clinton's second term coincided closely with the Phillips curve that had prevailed a generation earlier, in the 1950s and 1960s. Figure 17 shows the two Phillips curves, separated by two generations that included two oil shocks and the embrace of the neo-liberal regime. In all these curves inflation is paired with the previous year's employment, both theory and the data suggesting this lag structure.

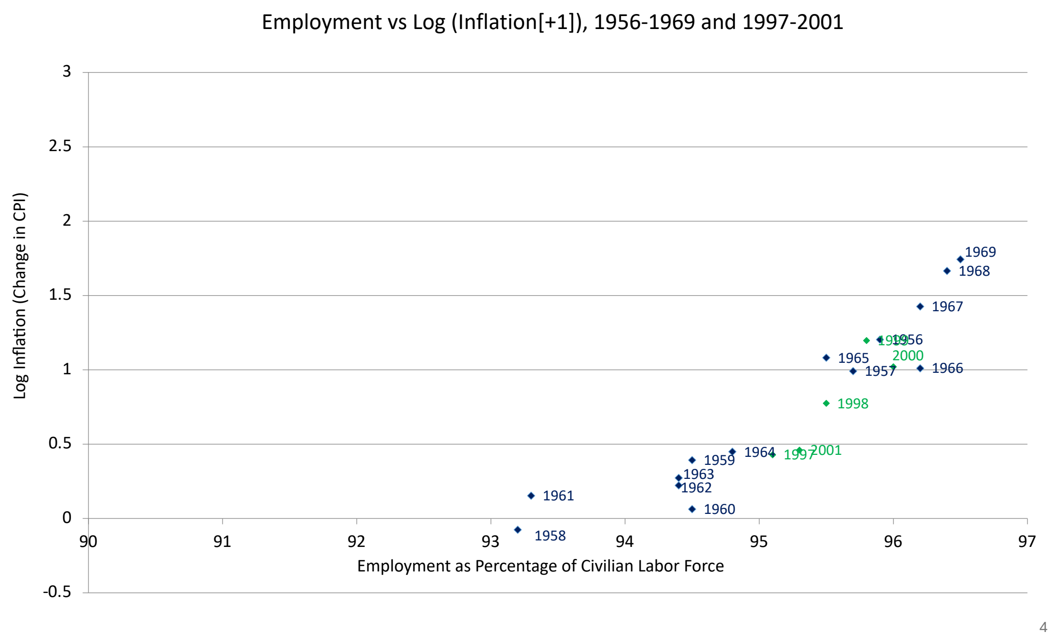


Figure 17

For the present exercise the rate of growth of GDP replaces employment as the measure of economic activity. As in the earlier analysis a lag of one year is assumed in relating real activity to inflation. The individual country experiences are summarized in the following table.

Table 3. Inflation Regressed on Lagged GDP Growth								
Country	China	China	France	Japan	Korea	Spain	UK	USA
Coefficient	0.995	1.379	0.329	0.327	0.526	0.644	0.230	0.211
t-statistic	3.84	3.06	3.15	3.68	5.37	5.09	1.26	2.62
Constant Term	-4.174	-7.259	0.873	-0.522	0.466	1.650	2.171	1.616
R2	0.39	0.40	0.32	0.29	0.40	0.38	0.04	0.14
No. of Observations	48	33	33	33	33	33	33	33
Interval	1971-2019	1986-2019	1986-2019	1986-2019	1986-2019	1986-2019	1986-2019	1986-2019

Except for China, observations begin 1986, after the stagflation caused by two oil shocks in the 1970s and the recovery in the early 1980s had run its course.

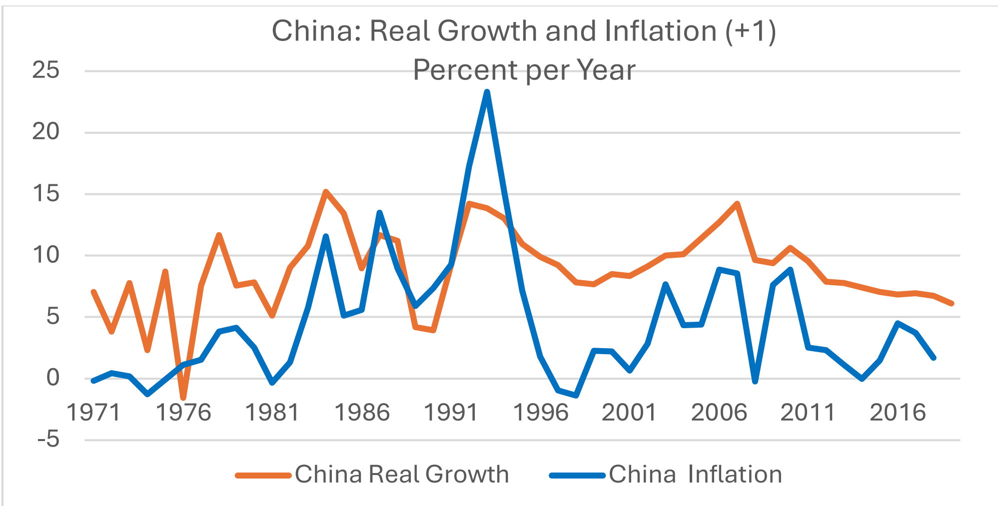


Figure 18

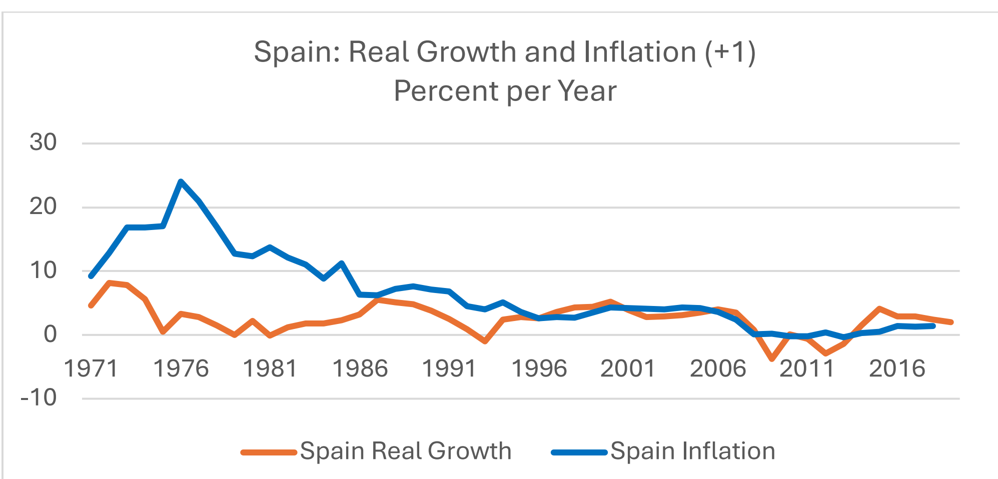


Figure 19

In my seven-country sample China is unique in escaping the oil-shock stagflation of the 1970s, as indicated by the comparison with Spain in Figures 18 and 19. The reason is simple. In the 1970s the role of oil in China's energy mix was minimal, 20 percent of its energy use; the other six countries ranged from just below 50 percent (the US and the UK) to over 70 percent (Japan). Consequentially, for China the relationship between inflation and real growth is much the same in the period before 1986 as afterwards, dominated by demand. (Even today, China is less reliant on oil in terms of the role of oil in its energy mix than the other countries in the sample, despite being the world's largest importer of oil.)

Observe that the only country that does not show a statistically significant relationship between real growth and inflation (at the 95 percent level) is the United Kingdom. As Figure 20 shows, from the mid '80s on the British economy had too little variation in both dimensions to establish a relationship between the two variables.

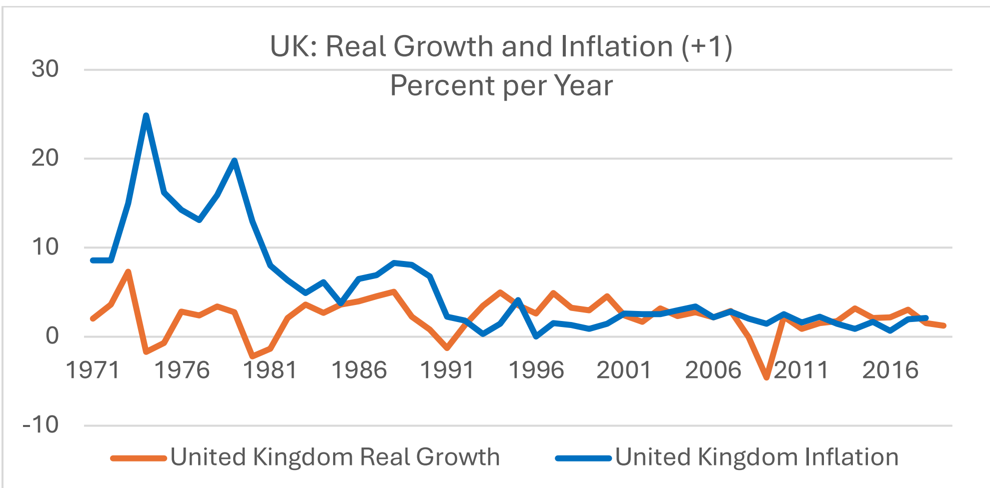


Figure 20

Conclusions

Where fundamentalists understand Keynes to have uncovered a fatal flaw in the invisible hand, the mainstream sees only a more sophisticated version of sand in the wheels. Both sand in the wheels—departures from the competitive model—and the fatal flaw—the failure of interest rates spontaneously to adjust aggregate demand to full-employment supply—can lead to unemployment in the short run, when the labor force is fixed exogenously. So, despite the analytic differences, both mainstream Keynesians and fundamentalists agree on the need for the visible hand of the government, first to nudge interest rates down and, in more extreme cases, to supplement monetary policy with activist fiscal policies.

The long run is another matter. Here the differences are not only theoretical. Mainstream macro sets up a wall between the supply side and the demand side; the frictions and rigidities that mar capitalism in the short run disappear sufficiently with time that the competitive model is an adequate guide to policy. And for the mainstream, the competitive market is a self-regulating miracle that ensures the efficient use of all available resources.

Supply-side variables determine real output, employment, growth, while demand affects only the price level. Interest rates balance investment demand and saving supply; capital formation will increase if either investment demand or the supply of saving increases. But the impact of investment and saving propensities is transitory: diminishing returns ensure that the capital-stock growth rate will sooner (rather than later) adjust to the "natural" rate determined by supply-side considerations. Government spending and taxation, fiscal policy, can also affect the composition of output and the rate of growth temporarily, but is no more effective in the long run than is monetary policy.

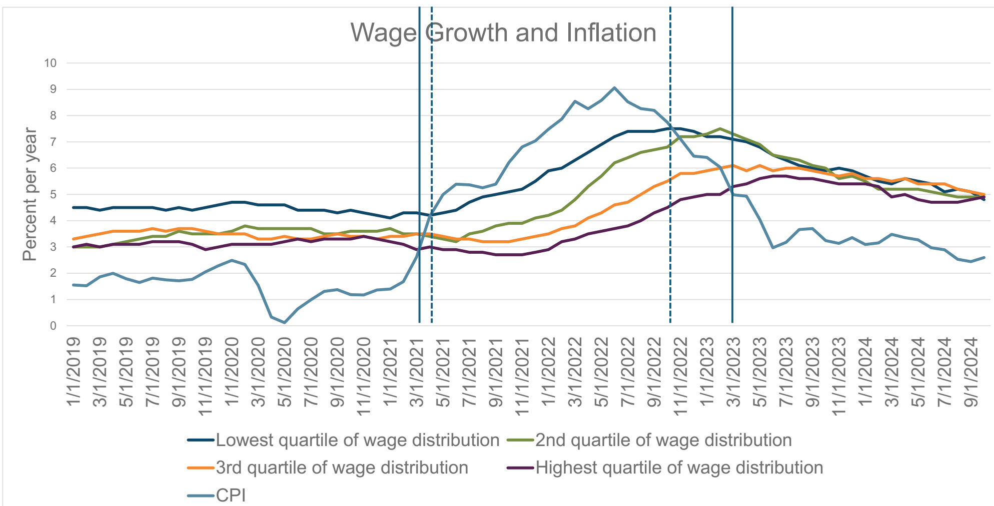
Among other uses, this theory is invoked to disarm opposition to central-bank independence. Since the central bank can have no lasting impact on output, employment, and growth, all that is left is to focus on inflation. "Inflation targeting" becomes a technocratic exercise akin to determining the weight limit of a bridge, rather than a political decision weighing the benefits of higher output, employment, and growth against the benefits of more stable prices.

"Political" is intended to emphasize the conflicting interests at play. For Martin Luther King Jr, the conflict was easily resolved. In *A Freedom Budget for All Americans,* published exactly 60 years ago, he and his co-signatories were clear:

> It would be a monstrous distortion of our values as a nation and a people to argue that we should balance the desirability of reducing unemployment (and meeting the other priorities of our national needs) against the prospects of some increases in the price level. (pp 74-75)

The *Freedom Budget* suggests that the beneficiaries of stable prices are all people with plenty to spare. And, yes, the bond-owning class, the folks who benefit most from stable prices, are indeed major beneficiaries of price stability. But not the sole beneficiaries. In the US, millions of retired schoolteachers, firemen, policemen and other state and municipal employees rely on fixed-income pensions that do not come with the robust cost-of-living adjustments that characterize the national social-security system. When inflation reaches levels that characterized the post-COVID years, their pocketbooks are especially hard hit.

Retirees are not the only non-wealthy to take a hit from inflation. And here the benefit:cost analysis gets tricky because the pain of inflation may be spread very unevenly. During the post-COVID recovery the top quartile of workers suffered significantly more than the bottom quarter. Not only were wage increases below price increases for a longer time (2 years vs 18 months), but the gap between the pace of price inflation and wage inflation was much larger. Figure 21 gives the picture.



Source: Atlanta Fed

Figure 21

In addition, the employment gains that accompanied inflation most likely favored the folks on the bottom. Grist perhaps for MLK's mill, but harder to digest than a simple comparison between fat cats losing from inflation and poor kittens benefiting from job gains.

This is only one of the political choices that emerges from a fundamentalist reading of Keynes. Once we recognize the possibility of choice, indeed the need for choice, we open a large box, including the resource constraints noted earlier in this essay.

Beyond resource constraints there are questions of the roles of both work and consumption in human fulfillment: does more of either mean better lives? This question has become all the more salient in a world in which AI is not only a promise but also a threat.

My purpose in this essay is not to answer these questions. It would take another essay to attempt even partial answers. My intention is rather to open debate, to convince you that long-run growth is about more than how quickly technology advances and policy about more than holding the line on prices.